Benoit Kokouvi NOUGNANKE

PhD Student in Computer Science

Education

2017-2021 Toulouse, France
Preparation of a PhD. Thesis in Computer Science.
Towards ML-based Management of Software-Defined Networks
* Advisor : Pr. Yann LABIT
* Keywords: SDN, Network management, Monitoring, Data centers, Machine Learning, Performance Optimization, Self-Driving Networks

Université Paul Sabatier / Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS-CNRS)
* Defense Date : 12/07/2021

2012-2017 Khouribga, Morocco
Engineer Degree in Computer Science – Networking and Telecommunications.
Ecole Nationale des Sciences Appliquées de Khouribga (ENSA)

Certification

2018
Machine Learning by Stanford University on Coursera.
Certificate earned at Sunday, June 3, 2018 2:16 PM GMT (Lien du certificat)

Summer School

2018
ResCom 2018: Data science, Network Science and Machine Learning.
Porquerolles, 18 - 22 June 2018

Professional Experiences

2020-2021 Toulouse, France
Teaching Assistant (ATER) at INSA Toulouse.
Programming, Operating Systems, UNIX, Introduction to Networking, Socket Programming, Openstack, Docker

August-September 2019 Tokyo, Japan
IIJ Innovation Institute.
International PhD Mobility
* Advisor : Marc BRUYERE

2017-2020 Toulouse, France
Université Paul Sabatier.
Teaching Assistant
* Computer Networks, Information Theory, Programming

March-August 2017 Toulouse, France
Laboratoire d'Analyse et d'Architecture des Systèmes (LAAS-CNRS).
Engineering Intern
* Implementation of a Software Defined Wireless Multi-hop Networking (SDWMN) Platform

September 2016 khouribga, Morocco
OCP - Khouribga (*Engineering Intern*) .
* Study of TETRA Standard

July-August 2016 Mohammédia, Morocco
CSI - Maroc (*Engineering Intern*).
* Implementation of A File Transfer Server and A Web Client

Grants

2020 Travel Grant ACM SOSR 2020

2019 International PhD Mobility Grant – Ecole Doctorale Systèmes (EDSYS)

2017 Ph.D. fellowship from French government (2017 - 2020)

2012 5 years Engineering Study Scholarship from Moroccan Agency for International Cooperation

Skills

Programming languages	Python, C, C++, P4, Matlab, R, Julia
Networking	SDN, Data center networking, OpenFlow, OpenVSwitch, Ryu, Mininet, NS3
Machine Learning	Scikit-learn, Keras, Tensorflow, Pytorch, Gym

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Other Skills	Time Series, Docker, Kubernetes, OpenStack
Languages	French (First Language), English (Fluent)

Publications

Conferences

2021 **ML-Based Incast Performance Optimization in Software-Defined Data Centers**.
Kokouvi Benoit Nougnanke, Yann Labit, Marc Bruyère
22nd International Conference on High-Performance Switching and Routing (IEEE HPSR 2021) – 7-10 June 2021. Paris, France.

2021 **Learning-based Incast Performance Inference in Software-Defined Data Centers**.
Kokouvi Benoit Nougnanke, Yann Labit, Marc Bruyère, Simone Ferlin, Ulrich Aïvodji
24th Conference on Innovation in Clouds, Internet and Networks (ICIN 2021) – March 01-04, 2021, Paris, France. **Best Paper Award**

2020 **Low-Overhead Near-Real-Time Flow Statistics Collection in SDN**.
Kokouvi Benoit Nougnanke, Marc Bruyère, Yann Labit
IEEE NEtSoft Conference on Network Softwarization 29 June-3 July 2020 – Virtual Conference

2020 **Novel Adaptive Data Collection based on a Confidence Index for Continuous Monitoring in SDN**.
Kokouvi Benoit Nougnanke, Yann LABIT
IEEE 17th Annual Consumer Communications & Networking Conference (CCNC 2020) – Las Vegas, USA

2017 **A Generic And Configurable Topology Discovery Service For Software Defined Wireless Multi-Hop Network**.
Lunde Chen, Slim Abdellatif, Pascal Berthou, Kokouvi Benoît Nougnanke, Thierry Gayraud
15th ACM International Symposium on Mobility Management and Wireless Access – Miami, FL, USA – November 21 - 25, 2017

Poster

2020 **Low-Cost Near-Real-Time Counters Collection in SDN**.
Kokouvi Benoit Nougnanke, Marc Bruyère, Yann Labit – ACM SOSR 2020

2018 **Control Messages Optimization in SDN**.
Kokouvi Benoit Nougnanke and Yann Labit – SDN DAY 2018 – Paris, France

Reports

2017 **Application Driven Networking in Dynamic Environments Cas d'application communications dans un théâtre d'opérations**.
Slim Abdellatif, Pascal Berthou, Benoit Nougnanke – LAAS Report - ANR Project

Presentations

2020 **BSFQ: Buffer Sizing and Fair Queuing to Rescue TCP Incast**.
SARA Meeting, Toulouse, France

2019 **Novel Adaptive Data Collection based on a Confidence Index for Continuous Monitoring in SDN Environments**.
IIJ Innovation Institute Summer Camp, Laforet Club Hakone Gora, Japan

2019 **Novel Adaptive Data Collection based on a Confidence Index for Continuous Monitoring in SDN Environments**.
STORE - ENSEEIHT, Toulouse, France

Internship Supervision

March-September 2021 **LY Yimtchen**.
Learning-based Traffic Performance Optimization in Software-Defined Data Centers

April-September 2019 **Sidoine KAMBIRE**.
Intelligent management of energy consumption of computer networks in SDN

July 2018 **Zahira TALEB**.
MANET networks on ns-3: Deployment and traces generation

Interests

Football, FIFA/PES, Music

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